Data management Assignment 1

2025

In this individual course assignment, your task will be to look at different stages of the database development project in case of a hypothetical case company, and perform different tasks that typically occur in a similar situation. Submit one file with your solution to Moodle. Please be aware that the submitted file will be checked in the plagiarism detection software after submission. The deadline is May 11, 23:59.

Background

In this assignment, the focus is on the data management challenges of a hypothetical online education company, LearnHubAI, that provides both live and self-paced courses to a global audience. The goal is to enhance data processing and management to support better learning analytics, operational decision-making, and improved student engagement. In the following, you assume the role of a newly hired data analyst. The company has tasked you with reviewing and proposing improvements to their internal data management systems.

Step 1: Formulating the business case (5 points)

Although LearnHubAI offers a wide range of online courses and has a growing user base, its internal data management practices are fragmented. Different teams collect data independently using spreadsheets, isolated tools, and separate cloud-based systems. As a result, many business decisions are based on incomplete or delayed information. LearnHubAI has identified three strategic priorities: (i) improving student engagement through better tracking of activity and performance; (ii) streamlining content management by integrating instructor and course data; and (iii) enhancing operational reporting across departments, especially in marketing and customer support.

Your task in this step is the following:

- Task 1: Draft a 2-page memo to the management explaining the limitations of LearnHubAI's current data management systems. Outline: (i) the advantages of a unified data management system, (ii) the types of data essential for operational and strategic decisions (e.g., user activity, enrollment patterns, quiz results, content ratings), and (iii) the key steps to transition to an integrated database system using a systems development life cycle approach.

Step 2: Conceptual design (15 points)

Following your memo, management has approved the development of a new database system. After discussions with business stakeholders, the following entities have been identified as relevant for supporting core business needs:

- Students: individuals who enroll in and complete courses. Attributes include Student ID, Name, Email, Country, and Account Type.
- Courses: educational content offered on the platform. Attributes include Title, Description, Category, Instructor, and Language.
- Instructors: content creators and facilitators. Attributes include Name, Area of Expertise, Email, and Country.
- Enrollments: records of which students are registered in which courses. Attributes include Course, Student, Enrollment Date, and Completion Status.
- Assessments: quizzes or assignments linked to specific courses. Attributes include Assessment, Course, Type, Max Score.
- Results: performance records from completed assessments. Attributes include Assessment, Student, Score, Submission Time.

In addition to the primary entities, the following data aspects are relevant. You may treat them as entities, attributes, or relationships depending on your design:

- Content Feedback: qualitative and quantitative student evaluations of courses.
- Login Activity: timestamps and metadata related to user logins and session durations.
- Course Versions: details on course updates and archival of previous materials.

Based on this information, you need to perform the following tasks:

- Task 1: Based on the goals outlined above, consider if any additional entities, attributes, or relationships should be captured in the design. Reflect on what LearnHubAI may want to track in the future.

- Task 2: Create an ER-diagram based on the provided and suggested specifications. Clearly explain your design choices, including identifiers and how relationships are constructed. Provide rationale for relationship type and cardinality.
- Task 3: Additionally, ensure that your final data model addresses these considerations:
  - Accommodating updated courses without losing historical data
  - Capturing student engagement and session duration
  - Including essential assessment data and feedback ratings
  - Supporting instructor performance review and content improvement processes

Step 3: Logical design (10 points)

After approving the conceptual model, management wants to move forward with a logical design. They are leaning toward a relational model but are also interested in understanding if other options like NoSQL could better support some aspects of their system—especially storing multimedia content and unstructured feedback.

- Task 1: Write a one-page summary comparing relational and non-relational databases. Discuss scenarios where NoSQL solutions may be more appropriate for LearnHubAI, particularly regarding scalability, feedback storage, and session tracking. Should the company consider a hybrid architecture?

As the decision to proceed with a relational design has been made, your main tasks in this phase are as follows:

- Task 2: Convert your ER diagram into a relational schema. Identify all primary and foreign keys. Define referential integrity constraints.
- Task 3: Evaluate the current normalization of your design. Identify which normal form your schema adheres to, and if needed, make adjustments to achieve at least third normal form.

Step 4: Physical database design and performance (10 points)

After the logical schema is completed, management is curious about techniques to ensure smooth performance. You suggest that denormalization might be beneficial in certain areas, especially those involving frequent queries or reporting.

- Task 1: Write a 1-page summary explaining the concept of denormalization. Identify 2–3 parts of your schema where denormalization could simplify data retrieval based on the typical use scenarios of the system. Consider reporting use cases like monitoring student progress, generating instructor dashboards, and analyzing course drop-off rates.

Task 2: conduct a usage analysis based on LearnHubAI's operational size.

- 50,000 registered students
- 2,000 available courses
- 400 instructors
- Each student completes 3 courses/year
- Each course contains 4–6 assessments
- Login activity is logged daily
- Course feedback is collected after each completion

Estimate the number of queries for: (i) course progress tracking, (ii) instructor performance reporting, and (iii) engagement pattern analysis. Based on this, reassess your denormalization proposals.

Step 5: Data Warehousing and data integration (5 points)

LearnHubAI currently has separate data marts for learning progress, content usage, and financial reporting. They are now considering a transition to a full Enterprise Data Warehouse (EDW) structure to centralize analytics.

- Task 1: In max. 1 page, describe the benefits of EDW compared to independent data marts. In the case of LearnHubAI, is near real-time data integration necessary for decision-making (e.g., triggering interventions based on student behavior)?

As the last task, in order to control data quality, you need to perform a basic check:

- Task 2: Consider your database model and identify attributes that are most likely to require manual input by staff or instructors. Which of these cannot be easily automated or standardized through dropdown menus, API connections, or data pipelines?

Step 6: Exploring the Role of LLMs in Data Management (5 points)

As a final task, you are asked to explore the potential role of Large Language Models (LLMs) in modern data management systems. To provide the background, you are expected to read and utilize the two provided academic articles: 'How Large Language Models Will Disrupt Data Management' and 'LLM-Enhanced Data Management'.

Task 1: Write a 1-2 page report in which you discuss the following topics:

- In what ways can LLMs support data management activities such as query generation, data cleaning, conceptual modelling etc.?
- What are the potential benefits of using LLMs in a company like LearnHubAI, especially in terms of improving data quality, decision-making, and automation?
- What are the challenges and risks, including issues related to accuracy, cost, explainability, or integration with existing systems?
- Reflect on your own database design in earlier steps (Step 2–5): would it be realistic or beneficial to integrate LLM-based components into any part of that architecture? Why or why not?

In your report, you are expected to refer to the two articles provided. In addition, you may draw on your own experience working with LLMs (e.g. ChatGPT or other tools), and incorporate insights from other academic or industry sources, provided they are properly cited.